JAMES HARRISON

Data Engineer (Staff Level)

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PROFESSIONAL SUMMARY

Experienced Data Engineer with 10 years building and maintaining large-scale data infrastructure across fintech and healthcare sectors. I've led teams through complex migrations, designed systems handling petabytes of data, and spent the last three years mentoring junior engineers while reducing operational costs by 40 percent. I prefer solving real problems over implementing trendy frameworks and take pride in writing code that's maintainable by others.

WORK EXPERIENCE

Senior Data Engineer

DataFlow Systems Ltd

Cambridge, United Kingdom

www.dataflowsystems.io

January 2021 - Present

Led architecture and implementation of real-time data pipeline processing 2.5 petabytes annually for EMEA clients across financial services and insurance sectors.

Designed and deployed Kubernetes-based infrastructure for 12 data engineers, reducing deployment time from 45 minutes to 8 minutes and cutting cloud infrastructure costs by 38 percent through optimized resource allocation.

Established data quality framework and monitoring using Apache Spark and custom Python validators, catching data anomalies within 15 minutes of ingestion rather than during reporting phase.

Mentored four junior engineers to mid-level capability, with two promoted to senior roles, implementing pair programming and weekly architecture reviews as standard practice.

Staff Data Engineer

Pinnacle Analytics

London, United Kingdom

www.pinnacle-analytics.co.uk

July 2017 - December 2020

Architected batch and streaming data platform using Apache Spark and Kafka for hedge fund clients, ingesting market data from 47 exchanges with guaranteed sub-second latency.

Migrated legacy ETL processes from Shell scripts and Python 2 to modern Scala and Apache Airflow, reducing manual interventions by 85 percent and enabling autonomous overnight runs.

Built self-service analytics platform allowing 60+ business stakeholders to query petabyte-scale datasets without SQL knowledge through drag-and-drop interface using Presto and Superset.

Investigated and resolved critical production incidents affecting portfolio calculations, including implementing circuit breakers and fallback mechanisms that prevented estimated 2.3 million pounds in trading losses.

Data Engineer

QuantumInsight Solutions

Bristol, United Kingdom

www.quantuminsight.net

September 2013 - June 2017

Developed and maintained Python-based data ingestion framework processing feeds from 23 different data providers, standardizing inconsistent schemas and formats into unified warehouse schema.

Implemented partitioning and indexing strategies on PostgreSQL and Cassandra clusters, improving query performance by average 64 percent and enabling real-time dashboards for 15 concurrent users.

Created automated testing suite and CI/CD pipeline using Jenkins and Python unittest, reducing production bugs by 71 percent and allowing safe daily deployments.

Collaborated with business analytics team to define and implement dimensional data model serving 120 business reports previously built in fragmented spreadsheets.

EDUCATION

University of Cambridge

Computer Science, BSc Honours

Cambridge, United Kingdom

2009 - 2012

TECHNICAL SKILLS

Programming Languages: Python, Scala, SQL, Java, Bash

Big Data Technologies: Apache Spark, Apache Kafka, Apache Airflow, Hadoop

Data Warehousing: PostgreSQL, Cassandra, Presto, Snowflake, Redshift

Cloud Platforms: AWS (EC2, S3, Lambda, RDS), Google Cloud Platform

Containerization and Orchestration: Docker, Kubernetes, Helm

Data Quality and Testing: Great Expectations, dbt, pytest, automated validation frameworks

Version Control: Git, GitHub Enterprise

Monitoring and Observability: Prometheus, Grafana, ELK Stack, DataDog

PROFESSIONAL CERTIFICATIONS

AWS Certified Data Analytics - Specialty, 2019

Cloudera Certified Associate Data Engineer, 2016